NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

MIL-PRF-6106/33C(USAF)
13 November 1987
SUPERSEDING
MIL-R-6106/33B(USAF)
16 September 1985

PERFORMANCE SPECIFICATION SHEET

RELAYS, ELECTROMAGNETIC, TYPE I, 400 AMPERES, WITH 5 AMPERE AUXILIARY CONTACTS, PERMANENT MAGNET DRIVE, HERMETICALLY SEALED

This specification is approved for use within the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the relays described herein shall consist of this specification and the latest issue of MIL-PRF-6106.

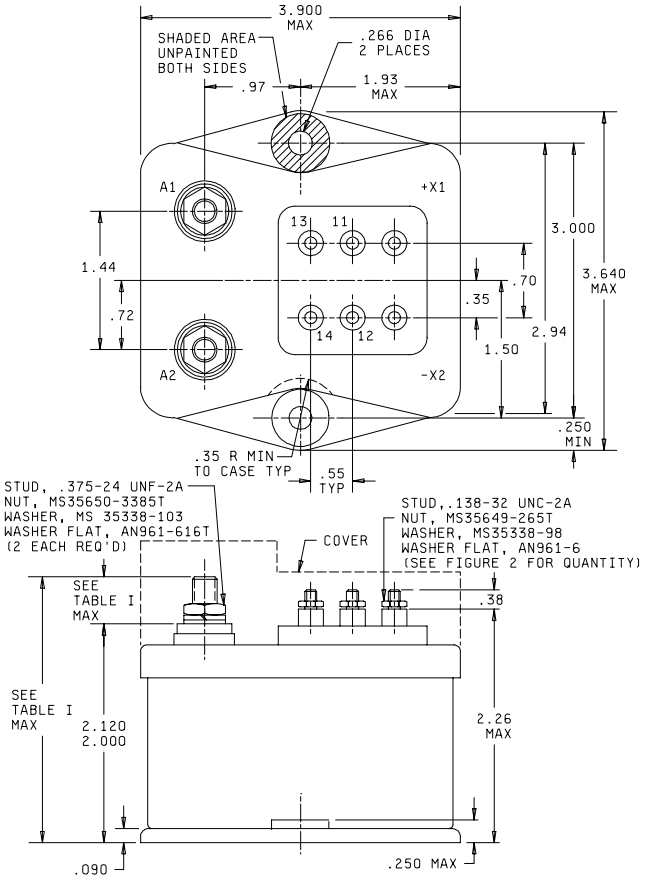


FIGURE 1. Dimensions and configurations.

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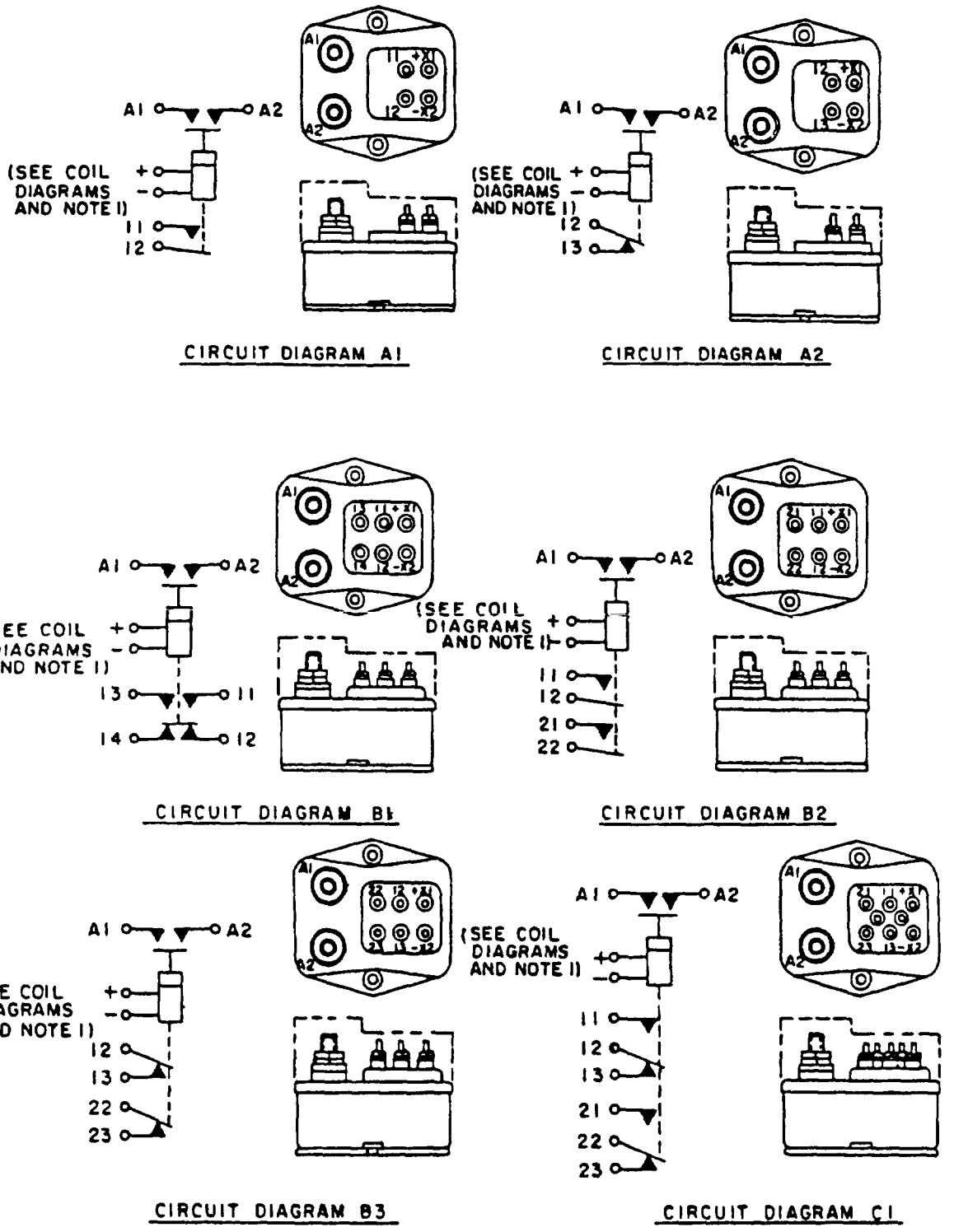
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Inches	mm	Inches	mm
.090	2.29	.97	24.6
.138	3.51	1.44	36.6
.250	6.35	1.50	38.1
.266	6.76	1.93	49.0
.33	8.4	2.000	50.80
.35	8.9	2.120	53.85
.375	9.53	2.26	57.4
.38	9.65	2.94	74.7
.55	14.0	3.000	76.20
.70	17.8	3.640	92.46
.72	18.3	3.900	99.06

NOTES
1 Dimensions are in inches.
2 Metric equivalents are given for general information only.
3. There shall be affixed to the relay a suitable legible circuit diagram that identifies each terminal location specified.
4 Permanent magnet drive consists of a permanent magnet with its flux path switched and combined with the electro-magnet flux.
5 Unless otherwise specified, tolerance is ±.010 (0.25 mm) for three place decimals and ±.03 (0.8 mm) for two place decimals.

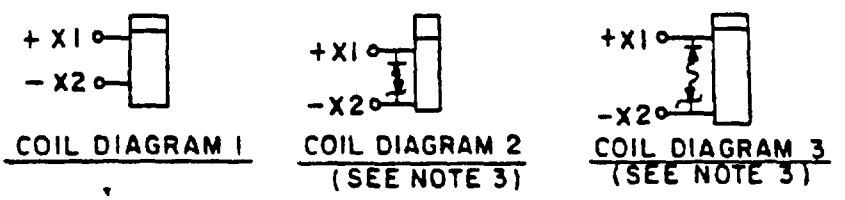
FIGURE 1. Dimensions and configurations - Continued.

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© FIGURE 2. Circuit diagrams and terminal configurations.

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NOTES:
1. To identify circuit diagrams, coil diagrams, and terminal configurations, see table I.
2. Number of terminals varies dependent upon number of auxiliary contacts.
3. Semiconductors shall be JANTX or JANTX screened. Coil diagram 3 shall continue to operate should the suppression circuit be in a failure mode (not applicable to coil diagram 2). Diodes shall have a peak inverse voltage rating of 600 V dc minimum.

© FIGURE 2. Circuit diagrams and terminal configurations - Continued.

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REQUIREMENTS:

Dash numbers and physical characteristics: See table I.

Duty rating. Continuous.

Coil data: See table II.

Operational data: See table II.

Contact data: See tables III and IV.

Contact voltage drop.

Initial: 0.150 volt.

After life: 0.175 volt.

Overload current: 2,400 amperes dc.

Rupture current: 3,000 amperes dc.

Interrupt current: Carry 2,000 amperes for 7 seconds and break five times (30 minutes off per cycle).

ELECTRICAL CHARACTERISTICS:

© Insulation resistance, initial: 100 megohms at 500 V dc.

After life or environmental tests: 50 megohms.

Dielectric withstanding voltage (sea level).

	Initial	After life tests
Coil to case	1,250 V rms	1,000 V rms
Auxiliary contacts	1,250 V rms	1,000 V rms
All other points	1,500 V rms	1,250 V rms

Dielectric withstanding voltage (altitude).

	50,000 feet
Coil to case	500 V rms
Auxiliary contacts	500 V rms
All other points	700 V rms

Physical:

Dimensions and configuration: See figure 1.

© Weight (maximum): 1.75 pounds.

© Terminal type: See figure 1, figure 2, and table II.

Terminal strength: Applicable.

Seal: Hermetic.

© Finish: Painted.

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© TABLE I. Mechanical and physical characteristics.

Dash numbers	Coil suppression (back EMF)	Circuit diagram	Coil diagram	Cover height dimensions	Terminal lengths
040	Not applicable	B1	1	3.38	0.75
041	Applicable	"	3	"	"
042	Applicable	"	3	"	"
043	Not applicable	"	1	"	"
044	Applicable	B2	3	"	1.13
045	Not applicable	B2	1	"	1.13
046	Applicable	A1	3	"	0.75
047	Applicable	A2	3	"	"
048	Not applicable	A1	1	"	"
049	Not applicable	A2	1	"	"
050	Applicable	B2	3	"	"
051	Not applicable	B2	1	"	"
052	Applicable	B3	3	"	1.13
053	Not applicable	"	1	"	1.13
054	Applicable	"	3	"	0.75
055	Not applicable	"	1	"	0.75
056	Applicable	C1	3	Not applicable	1.13
057	Not applicable	"	1	"	1.13
058	Applicable	"	3	"	0.75
059	Not applicable	"	1	"	0.75
060	Applicable	B2	2	3.38	1.13
061	Applicable	B3	"	3.38	"
062	"	C1	"	Not applicable	"
063	"	B1	"	3.38	0.75
064	"	"	"	"	0.75
065	Not applicable	"	1	"	1.13
066	Applicable	"	3	"	"
067	"	"	2	"	"
068	"	B2	"	"	0.75
069	"	A1	"	"	0.75
070	Not applicable	"	1	"	1.13
071	Applicable	"	3	"	"
072	"	"	2	"	"
073	"	A2	2	"	0.75
074	Not applicable	"	1	"	1.13
075	Applicable	"	3	"	"
076	"	"	2	"	"
077	"	B3	"	"	0.75
078	"	C1	"	Not applicable	0.75

© TABLE II. Operating characteristics

Coil data											Time (milliseconds maximum)						
	Nominal		Maximum		Max pickup voltage			Dropout	Hold	Back			Contact bounce				Break
													Main		Auxiliary		
Coil diagram	Volts dc	Res. ±10% at 25°C	Volts dc	Ampere	Nominal at 71°C	High temp test	Cont. current test	voltage 1/ (√dc)	voltage 1/ 2/ (√dc)	EMF (V dc max)	Operate	Release	NO	NC	NO	NC	bounce NO only
1	28	60	32	.6	18	19	21	1.5	9.0	NA	35	15	4	NA	4	4	NA
2 and 3	28	60	32	.6	18	19	21	1.5	9.0	42	35	15	4	NA	4	4	0.1

1/ Over temperature range

2/ Must pass random vibration test using a 9 V dc hold voltage 25°C.

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© TABLE III. Rated contact load endurance (ampere per pole) case grounded.

Type of load	(Endurance) life operating cycles x 10^3	28 V dc Main		28 V dc/115 400 Hz auxiliary	
		NO	NC	D.B.	S.B.
Resistive	50	400	---	5	5
Inductive	20	150	---	5	3
Motor 1/ 2/	50	250	---	.	
Lamp				1	1
Mechanical life					
Reduced current	100	100			
Intermediate current	Not applicable				

1/ Off time for motor load test shall be 6 seconds maximum.

2/ Shall also meet 20,000 cycles of special motor load as specified in table IV.

TABLE IV. Motor load (special). 1/ 2/

Elapsed time seconds	Starter current amperes
0 (initial)	1,200
1	1,000
2	1,000
3	900
4	800
5	700
6	600
7 to 60	400

1/ Must be able to meet the requirements specified for three cycles with only 10 seconds off between cycles. At the end of three cycles, off time will be 5 minutes, minimum.

2/ The start cycle may be aborted any time during this period, requiring a contact interrupt capability of at least 1,000 amperes.

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ENVIRONMENTAL CHARACTERISTICS:

Temperature rating: -55°C to +71°C.

Maximum altitude rating: 50,000 feet.

Shock G level: 25 G's.

Duration: 9 ms.

Maximum duration contact opening: 10 µs.

Vibration, sinusoidal:

G level:	10 G's	5 G's
Frequency range:	70-500 Hz	500-2,000 Hz.

Random: 1 hr/plane
0.04 G^2/Hz 15-197 cycles.
4 dB/octave increase.
197-300 cycles.
0.07 G^2/Hz 300-1,000 cycles.
6 dB/octave decrease.
1,000-2,000 cycles.

© Acceleration: 15 G's.

PART NUMBER: M6106/33- (dash number from table I).

© SUPERSESSION DATA: See table V.

TABLE V. Supersession data.

Superseded part number M6106/33-	New part number M6106/33-	Superseded part number M6106/33-	New part number M6106/33-
001	040	021	060
002	041	022	061
003	042	023	062
004	043	024	063
005	044	025	064
006	045	026	065
007	046	027	066
008	047	028	067
009	048	029	068
010	049	030	069
011	050	031	070
012	051	032	071
013	052	033	072
014	053	034	073
015	054	035	074
016	055	036	075
017	056	037	076
018	057	038	077
019	058	039	078
020	059		

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© Cross reference for Government logistics support: See table VI.

TABLE VI. Cross reference for Government logistics support.

Superseded part number M6106/33-	New part number M6106/33-	Support part number M6106/33-	Superseded part number M6106/33-	New part number M6106/33-	Support part number M6106/33-
001	040	040	021	060	060
002	041	041	022	061	061
003	042	042	023	062	062
004	043	043	024	063	063
005	044	044	025	064	064
006	045	045	026	065	065
007	046	046	027	066	066
008	047	047	028	067	067
009	048	048	029	068	068
010	049	049	030	069	069
011	050	050	031	070	070
012	051	051	032	071	071
013	052	052	033	072	072
014	053	053	034	073	073
015	054	054	035	074	074
016	055	055	036	075	075
017	056	056	037	076	076
018	057	057	038	077	077
019	058	058	039	078	078
020	059	059			

Custodian:
Air Force - 85

Review activities:
Air Force - 99
DLA - ES

Preparing activity:
Air Force - 85

Agent:
DLA - ES

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